

Advanced Econometrics

ARIMA

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Outline

- 1 Recap: Random Walk
- 2 AR, MA & ARMA Processes
- 3 Box-Jenkins Procedure
- 4 Real Data: S&P 500
- 5 Forecasting
- 6 Summary

Random Walk Without Drift

Intuition first

Imagine a drunk person taking steps: each step goes **randomly left or right**, but always *starting from wherever they currently are*.

Mathematically:

$$\underbrace{y_t}_{\text{today's position}} = \underbrace{y_{t-1}}_{\text{yesterday's position}} + \underbrace{\varepsilon_t}_{\text{random step}}, \quad \varepsilon_t \sim \mathcal{N}(0, 1)$$

ACF (*how correlated is today with the past?*)

- Decays *extremely slowly* — today is still correlated with observations from *months ago*
- Nearly **all lags stay significant**
- **This is the red flag for non-stationarity!**

PACF (*direct effect of each lag?*)

- **Only lag 1 matters** (≈ 1.0)
- All other lags ≈ 0
- Intuition: once you know yesterday, knowing the day before adds *nothing new*

How to spot a random walk: the two-second rule

ACF vs PACF — What Are We Actually Plotting?

Key insight

Same data. Same series. But two completely different questions.

ACF at lag 2

Just calculate plain correlation:

$$\text{corr}(y_t, y_{t-2})$$

Take series, shift by 2, correlate.

No tricks. No adjustments.

y_t	y_{t-2}
5	3
7	4
6	5
8	7
...	...

PACF at lag 2

Run a regression:

$$y_t = \beta_1 y_{t-1} + \beta_2 y_{t-2} + e$$

Run OLS. Take **only** β_2 .

That is your PACF bar at lag 2.

What β_2 means

“How much does y_{t-2} still help
after y_{t-1} **is already included?**”

If yesterday already explains
everything $\Rightarrow \beta_2 \approx 0$
 $\Rightarrow$ **PACF bar $\approx$ zero**

ACF vs PACF

	ACF	PACF
Formula	$\text{corr}(y_t, y_{t-k})$	β_k from OLS on lags $1 \dots k$
Measures	Total similarity	Direct effect only
Includes	Direct + indirect paths	Direct path only
Useful for	Spotting MA order	Spotting AR order
AR process	Decays slowly	Cuts off sharply
MA process	Cuts off sharply	Decays slowly

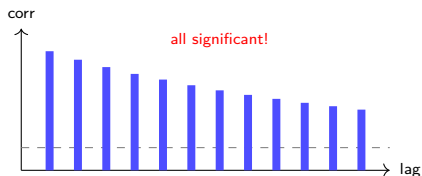
Height analogy

ACF: are grandfather and you correlated in height? **Yes — 0.5**

PACF: are grandfather and you correlated *after already knowing your father's height*? **Nearly 0** — father already carries all the information

Random Walk Without Drift

ACF (barely decays — all bars tall)



PACF (only lag 1 is big)



How to spot a random walk: the two-second rule

ACF barely moves (all bars tall) + **PACF drops after lag 1** $\Rightarrow$ **non-stationary. You must difference the series!**

Random Walk Without Drift

Definition

$$y_t = y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1)$$

ACF characteristics:

- Decreases *very slowly* (almost no decay)
- Nearly all lags remain significant

PACF characteristics:

- Only lag 1 is significant (≈ 1)
- All other lags near zero

Key takeaway

Slow ACF decay + single PACF spike $\Rightarrow$ non-stationary (random walk)!

AR(p) — Autoregressive Process

General Form

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

	ACF	PACF
AR(1)	Exponential decay	1 significant spike
AR(2)	Exponential decay	2 significant spikes
AR(p)	Exponential decay	p significant spikes

Example — AR(1): $y_t = 0.8 y_{t-1} + \varepsilon_t$

PACF lag 1 ≈ 0.8 (the autoregressive parameter).

What is an Autoregressive (AR) Process?

One sentence explanation

Today's value = weighted sum of your own past values + random surprise

"Auto" = self "Regressive" = regression $\Rightarrow$ **regressing on yourself**

AR(1) — the simplest case:

$$\underbrace{y_t}_{\text{today}} = 0.8 \cdot \underbrace{y_{t-1}}_{\text{yesterday}} + \underbrace{\varepsilon_t}_{\text{surprise}}$$

- If yesterday = 100
today $\approx 80 + \text{noise}$
- Next day $\approx 64 + \text{noise}$
- Slowly **pulls back to zero**
 $\rightarrow$ called **mean reversion**

The number p = how far back you look:

- AR(1)** only yesterday
- AR(2)** yesterday + 2 days ago
- AR(p)** last p days all matter

Real-life analogy — your mood:

- 60% depends on yesterday's mood
- 20% depends on 2 days ago
- rest = random (bad weather, coffee...)

$\Rightarrow$ That is **AR(2)**!

AR Process — Why the Parameter Value Matters

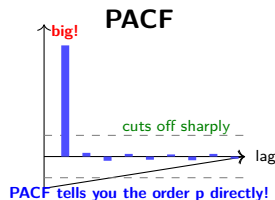
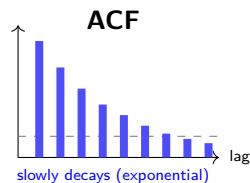
$$\text{AR}(1): y_t = \phi \cdot y_{t-1} + \varepsilon_t$$

ϕ	Behaviour	Stationary?
$ \phi < 1$	Mean reversion	Yes
$\phi = 1$	Random walk	No
$ \phi > 1$	Explodes	No

Key rule

All AR parameters inside the **unit circle** $\Rightarrow$ stationary
(for AR(1): just $|\phi| < 1$)

How to spot AR in plots:



ARMA(p,q) — Mixed Process

General Form

$$y_t = \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

	ACF	PACF
ARMA(p,q)	Exponential decay	Exponential decay

Identification challenge

Both ACF and PACF tail off — use **information criteria** (AIC/BIC) to select orders p and q .

Rule of thumb: Slower ACF decay than PACF $\Rightarrow$ AR effect dominates.

ARMA(p,q) — The Hybrid Model

Intuition first

ARMA = **AR** (depends on past *values*) + **MA** (depends on past *shocks*)

Think of it as: today depends on **where you were** AND **what surprised you**

ARMA(1,1) — simplest example:

$$\underbrace{y_t}_{\text{today}} = \underbrace{\phi_1 y_{t-1}}_{\text{AR part:}} + \underbrace{\varepsilon_t}_{\text{today's}} + \underbrace{\theta_1 \varepsilon_{t-1}}_{\text{MA part:}}$$

$\underbrace{\hspace{1.5cm}}$ $\underbrace{\hspace{1.5cm}}$ $\underbrace{\hspace{1.5cm}}$
 past value shock past shock

General ARMA(p,q):

$$y_t = \underbrace{\phi_1 y_{t-1} + \cdots + \phi_p y_{t-p}}_{\text{AR part: last } p \text{ values}} + \underbrace{\varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}}_{\text{MA part: last } q \text{ shocks}}$$

Real life example — your grade:

- AR part: last week's grade matters
- MA part: last week's surprise (exam was harder than expected) still affects you today

	ACF	PACF
AR only	decays	cuts off
MA only	cuts off	decays
ARMA	decays	decays

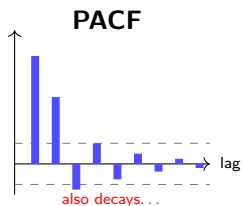
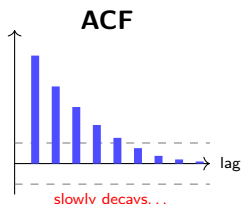
ARMA — The Identification Problem

The bad news

Both ACF and PACF slowly decay and **never cut off cleanly**

⇒ you cannot read off p and q directly from the plots!

What the plots look like:



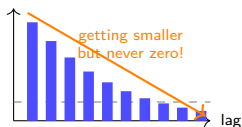
So what do you do?

- 1 Try several models
ARMA(1,1), ARMA(2,1),
ARMA(1,2), ARMA(2,2)...
- 2 Compare AIC and BIC
lower = better fit
- 3 Check residuals
must be white noise
- 4 Pick the winner

Quick Reference: ACF/PACF Patterns

Process	ACF	PACF	Stationary?
Random Walk	Very slow decay	Spike at lag 1 only	No
AR(p)	Exp. decay	Cuts off at lag p	Yes
MA(q)	Cuts off at lag q	Exp. decay	Yes
ARMA(p, q)	Exp. decay	Exp. decay	Yes

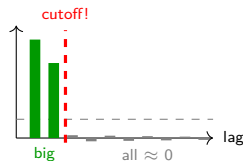
Decays = slow fade, never stops



Cannot read the order.

You are stuck — use AIC/BIC.

Cuts off = sudden wall, then zero



2 big bars then sudden zero

⇒ **order = 2. You can read it directly!**

The Box-Jenkins Procedure

1 Initial identification

- Test stationarity (ADF test)
- Determine d (order of integration)
- Inspect ACF/PACF to suggest p and q

2 Estimation

- Fit candidate ARIMA(p, d, q) models
- Check parameter significance (`coefstest()`)

3 Diagnostics (iterate with step 2)

- Residuals white noise? $\rightarrow$ ACF/PACF of residuals
- Ljung-Box test (`Box.test()`)
- Compare models: AIC vs. BIC

4 Forecasting

- Out-of-sample evaluation (MAE, MSE, MAPE)

Step 1 — Import & Inspect Data

```
library(quantmod)
getSymbols("^GSPC", from = "1970-01-01", to = "2017-03-29")

GSPC <- GSPC[, 6]           # Keep Adjusted Close
names(GSPC) <- "SP500"

GSPC$dSP500 <- diff.xts(GSPC$SP500)  # First differences

plot(as.zoo(GSPC), main = "Original and differenced S&P500")
```

Why first differences?

Stock prices are typically **I(1)**: the level is non-stationary but the first difference (daily returns) is stationary.

Step 1 — ADF Test (Integration Level)

```
source("function_testdf2.R")

# Level: with constant and trend
testdf2(GSPC$SP500, test.type = "ct",
        max.augmentations = 7, max.order = 5)

# Level: with constant only
testdf2(GSPC$SP500, test.type = "c",
        max.augmentations = 7, max.order = 5)

# First difference: no constant
testdf2(GSPC$dSP500, test.type = "nc",
        max.augmentations = 7, max.order = 5)
```

Result

Level: H_0 (non-stationary) **cannot** be rejected $\Rightarrow d \geq 1$

First diff: H_0 **rejected** $\Rightarrow d = 1$

Step 2 — ACF/PACF of First Differences

```
par(mfrow = c(2,1))
acf(GSPC$dSP500, lag.max = 36, ylim = c(-0.1, 0.1),
    lwd = 5, col = "dark green", na.action = na.pass)
pacf(GSPC$dSP500, lag.max = 36,
    lwd = 5, col = "dark green", na.action = na.pass)
par(mfrow = c(1,1))
```

What to look for

- Significant spikes at lags 1, 2, 5 in both ACF and PACF
- Lags 3 and 4 appear negligible
- Suggests ARIMA(5,1,5) *possibly without lags 3, 4*

Step 3 — Estimating ARIMA(1,1,1) Baseline

```
library(forecast)

arima111_2 <- Arima(GSPC$SP500,
                    order = c(1,1,1),
                    include.constant = TRUE)

summary(arima111_2)
coeftest(arima111_2)

# Residual diagnostics
Box.test(resid(arima111_2), type = "Ljung-Box", lag = 10)
```

Ljung-Box result

H_0 (white noise residuals) **rejected** — model is incomplete!

Autocorrelation remains in residuals $\Rightarrow$ need higher-order model.

Step 3 — ARIMA(5,1,5) Full Model

```

arima515 <- Arima(GSPC$SP500,
                  order = c(5,1,5),
                  include.constant = TRUE)
coeftest(arima515)

Box.test(resid(arima515), type = "Ljung-Box", lag = 10)
# H0 cannot be rejected -> residuals are white noise!

```

Restricted version — drop lags 3 & 4

```

arima515_2 <- Arima(GSPC$SP500, order = c(5,1,5),
                    fixed = c(NA,NA, 0, 0,NA, # AR: keep 1,2,5
                              NA,NA,NA, 0, 0,NA), # MA: keep 1,2,3
                    include.constant = TRUE)

```

Step 3 — Model Comparison (AIC / BIC)

```
AIC(arima111_2, arima515, arima515_2,
    arima515_3, arima510, arima015)

BIC(arima111_2, arima515, arima515_2,
    arima515_3, arima510, arima015)

# Automated search
arima.best.AIC <- auto.arima(GSPC$SP500, d=1,
    max.p=7, max.q=7, ic="aic",
    stepwise=FALSE, allowdrift=TRUE, trace=TRUE)
```

Model	Best AIC?	Best BIC?
ARIMA(5,1,5) full	✓	
ARIMA(1,1,1)		✓
auto.arima: (2,1,5)	✓	

Forecasting — Setup

```
# Hold out last 8 observations
GSPC.sample <- GSPC["/2017-03-17", 1]

# Re-estimate on shorter sample
arima515s <- Arima(GSPC.sample$SP500,
                   order = c(5,1,5),
                   include.constant = TRUE)

# Forecast 8 periods ahead
forecast515 <- forecast(arima515s, h = 8)
forecast515
```

Forecasting — Visualisation

```
plot(ts(GSPC[,1]), main = "8-day forecast of S&P500",  
      xlim = c(11880, 11919), ylim = c(2250, 2450))  
abline(v = 11912, lty = 2, col = "gray")  
lines(forecast515$mean, col = "red", lwd = 2)  
lines(11912:11919, forecast515$lower[,2],  
      col = "red", lty = 3)  
lines(11912:11919, forecast515$upper[,2],  
      col = "red", lty = 3)
```

Confidence intervals

forecast515\$lower[,1] = 80% lower; [,2] = 95% lower

forecast515\$upper[,1] = 80% upper; [,2] = 95% upper

Forecast Quality Metrics

```
SP500.r <- tail(GSPC$SP500, 8)           # real values
SP500.f <- as.numeric(forecast515$mean)   # forecasts

SP500_fc <- data.frame(SP500.r, SP500.f)
SP500_fc$mae <- abs(SP500.r - SP500.f)
SP500_fc$mse <- (SP500.r - SP500.f)^2
SP500_fc$mape <- abs((SP500.r - SP500.f) / SP500.r)
SP500_fc$amape <- abs((SP500.r - SP500.f) /
                      (SP500.r + SP500.f))
colMeans(SP500_fc[, 3:6])
```

Important!

The model with the **best in-sample fit** (lowest AIC/BIC) does **not** necessarily produce the best *out-of-sample* forecasts. Always compare forecast performance across models!

Key Takeaways

Identification rules

- ACF cuts off, PACF decays $\Rightarrow$ MA
- PACF cuts off, ACF decays $\Rightarrow$ AR
- Both decay $\Rightarrow$ ARMA
- Very slow ACF decay $\Rightarrow$ non-stationary, difference first

Model selection

- AIC tends to favour larger models
- BIC penalises complexity more
- Residuals *must* be white noise
- Automated (`auto.arima`) $\neq$ always best
- Forecast performance $\neq$ in-sample fit

Box-Jenkins in one line

Identify d (ADF) $\rightarrow$ **Identify** p, q (ACF/PACF) $\rightarrow$ **Estimate** $\rightarrow$ **Diagnose** $\rightarrow$ **Forecast**

ARMA — Two Ingredients, One Model

You already know:

AR = today depends on **past values**

$$y_{t-1}, y_{t-2}, \dots$$

MA = today depends on **past shocks**

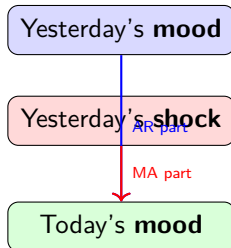
$$\varepsilon_{t-1}, \varepsilon_{t-2}, \dots$$

ARMA = just use **BOTH**:

$$y_t = \underbrace{0.8 y_{t-1}}_{\text{past value}} + \underbrace{\varepsilon_t + 0.5 \varepsilon_{t-1}}_{\text{past shock}}$$

That is it. Nothing more.

Mood analogy:



AR: happy yesterday $\Rightarrow$ happy today

MA: bad news yesterday
still drags you down today

ARMA: both at the same time

ARIMA — The Extra Letter I

Problem: ARMA only works on stationary series!

Stock prices, GDP, house prices — all trend upward forever

⇒ **non-stationary** ⇒ cannot fit ARMA directly

Solution: differencing

Instead of modelling the **level**:

100 102 99 105 103...

Model the **changes**:

+2 -3 +6 -2...

Differencing = subtract yesterday

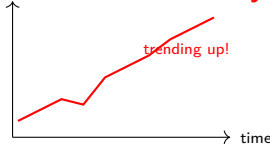
$$\Delta y_t = y_t - y_{t-1}$$

Changes bounce around zero

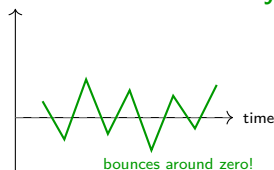
⇒ **stationary!**

S&P500 example:

Level — non-stationary



Differenced — stationary



ARIMA(p, d, q) — Three Numbers Explained

Letter	Meaning	Plain English	Example
p	AR order	how many past values you use	$p = 1$: use yesterday
d	Integration order	how many times you difference	$d = 1$: subtract yesterday
q	MA order	how many past shocks you use	$q = 1$: use yesterday's shock

ARIMA(1,1,1) step by step:

- 1 Stock price is non-stationary (trends upward forever)
- 2 **Difference once** ($d = 1$):
now you have daily returns
 $\Rightarrow$ stationary
- 3 **Fit ARMA(1,1)** on returns:
use 1 past value + 1 past shock

Special cases

- ARIMA(**1,0,0**) = plain AR(1)
(already stationary)
- ARIMA(**0,0,1**) = plain MA(1)
(already stationary)
- ARIMA(**0,1,0**) = random walk
(just difference it!)

How Do We Know If Our Forecast Is Good?

Intuition

After forecasting, compare **what you predicted** vs **what actually happened**.
The smaller the error — the better the model.

Example — 3 day forecast:

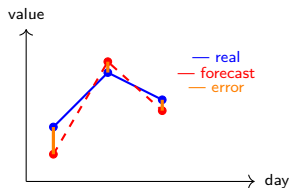
Day	Real	Forecast	Error
1	100	95	-5
2	200	210	+10
3	150	145	-5

Problem: errors cancel out!

$$-5 + 10 - 5 = 0$$

Looks perfect — but it is **not**!

⇒ We need better measures



MAE, MSE, MAPE — Three Ways to Measure Error

MAE — Mean Absolute Error

$$\text{MAE} = \frac{1}{n} \sum \underbrace{|y_t - \hat{y}_t|}_{\text{absolute error}}$$

Take each error, **ignore the sign**, then average.

Day 1: $|-5| = 5$

Day 2: $|+10| = 10$

Day 3: $|-5| = 5$

MAE = $(5 + 10 + 5)/3 = 6.7$

Pro: easy to understand

Con: treats all errors equally

MAPE — Mean Absolute % Error

$$\text{MAPE} = \frac{1}{n} \sum \left| \frac{y_t - \hat{y}_t}{y_t} \right| \times 100\%$$

Same as MAE but **as a percentage**

MSE — Mean Squared Error

$$\text{MSE} = \frac{1}{n} \sum \underbrace{(y_t - \hat{y}_t)^2}_{\text{squared error}}$$

Take each error, **square it**, then average.

Day 1: $(-5)^2 = 25$

Day 2: $(+10)^2 = 100$

Day 3: $(-5)^2 = 25$

MSE = $(25 + 100 + 25)/3 = 50$

Pro: punishes big errors more

Con: hard to interpret (units squared)

Which one to use?

MAE just want average error

MSE big errors are very bad

MAPE want a % — easy to explain

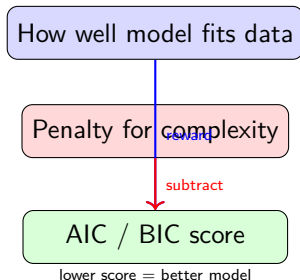
AIC and BIC — Choosing the Best Model

The problem

Adding more parameters **always** improves fit

— but the model becomes too complicated and fits **noise**, not the real pattern

⇒ We need to **penalise complexity**



	AIC	BIC
Penalty	smaller	bigger
Prefers	larger models	simpler models
Use when	prediction	explanation

Exam analogy:

Imagine two students:

Student A: memorised every detail — perfect on training data, fails on new questions

Student B: understood the concept — slightly worse on training, great on new questions

AIC and BIC — The Key Rules

Rule 1: lower = better

AIC = -120 is **better** than AIC = -80

BIC = 200 is **better** than BIC = 250

Always minimise!

Rule 2: only compare same data

AIC from model A on S&P500

cannot be compared to

AIC from model B on gold price

Must be **same dataset**!

Rule 3: AIC vs BIC disagree?

Report both.

Pick based on your goal:

forecasting $\rightarrow$ AIC

interpretation $\rightarrow$ BIC

Example — comparing 4 models:

Model	AIC	BIC
ARIMA(1,1,1)	-850	-845
ARIMA(2,1,2)	-860	-848
ARIMA(5,1,5)	-880	-830
ARIMA(1,1,0)	-820	-818

AIC picks: ARIMA(5,1,5)

bigger model, fits data better

BIC picks: ARIMA(1,1,1)

simpler model, avoids overfitting

Bottom line

MAE/MSE/MAPE = how good
are your **forecasts**?

AIC/BIC = which model
fits the **data** best?

You need **both**!